

Fabric Tag#: Envisioning a UXD Solution for Textile Culture Preservation and Fashion Sustainability

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Abstract

This paper explores Fabric Tag#, an innovative solution designed to address the environmental challenges posed by fast fashion while celebrating the cultural and industrial heritage of fabric. Focused on Tainan, once known as the "City of Fabric," the study investigates the local fabric market through field studies and persona development, revealing key insights for designing a sustainable, user-centered experience. Fabric Tag# integrates physical fabric tags with a digital app, providing users with a deeper connection to fabric through detailed documentation of its material, craftsmanship, and story. The app also supports local fabric retailers by offering market trend insights, inventory management, and a platform for sharing cultural narratives. By blending technology and sustainability, Fabric Tag# aims to foster a community around fabric appreciation, promote sustainable practices, and inspire users to rethink their relationship with textiles in an era dominated by fast fashion.

CCS Concepts

• **Human-centered computing** Interaction design Interaction design process and methods User centered design;

Keywords

Fashion Sustainability, Textile Cultural Preservation, Urban Culture Exploration, Physical and Digital Interaction

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1 Introduction

In the contemporary era of rapid consumerism, fast fashion has emerged as a dominant force in the market [1], characterized by its affordability and the swift turnover of styles [2]. However, beneath this seemingly irresistible allure lies a significant environmental challenge: the generation of immense fabric waste [3], which exacerbates global pollution. This issue underscores the urgent need to address the sustainability of the fashion industry [4].

Historically, fabric held a central role in everyday life [5], embodying cultural traditions, craftsmanship, and the beauty of human creativity. Each piece of fabric carried a story, connecting individuals to their heritage and surroundings [6]. Yet, in the relentless pursuit of fleeting trends, society has gradually lost sight of this cultural richness, reducing fabric to a mere disposable commodity [7].

The discourse around sustainable fashion [8], cultural preservation, and the revitalization of traditional industries has increasingly highlighted the need to reimagine our relationship with textiles. Fabric, as both a material and a symbol, bridges environmental concerns with cultural identity, industrial heritage, and innovative potential. These interconnections make it a pivotal element in addressing some of the most pressing challenges of our time.

Fabric Tag# aims to bring fabric into focus within the urban landscape. It encourages a paradigm shift—moving away from consumption and towards mindful appreciation. By celebrating and preserving the cultural narratives embedded in textiles, this initiative seeks to transform the way people perceive and interact with fabric. Through such efforts, we can safeguard cultural heritage, promote sustainability, and inspire a renewed sense of connection to the beauty and meaning in everyday life.

2 Problem exploration

Based on the issues surrounding fast fashion and textile culture, we discovered that our hometown, Tainan, was once a prominent textile hub in Taiwan [9]. During the 1980s, the Ximen Asakusa Fabric



Figure 1: Ximen Asakusa Fabric Market
(<https://www.facebook.com/photo.php?fbid=1110938037056327&id=100044204237762&set=a.669158284567640>)

Market thrived as a bustling center of fabric trade [10]. However, due to the transformation of the textile industry and the rise of fast fashion [11], [12], the number of fabric and tailoring shops has gradually declined [9], leaving the market as a destination primarily visited by those with specific needs for fabric purchases.

In response, our team conducted a series of studies on the Ximen Asakusa Fabric Market [13], aiming to design an experiential framework that promotes the value of textile culture. Through this approach, we seek to redefine the place of textile culture and local fabric businesses within the urban landscape, revitalizing their role and significance in contemporary society.

2.1 Field Studies and Generate Personas

To begin, our team conducted user observations at the Ximen Asakusa Fabric Market. Over three separate time periods, we observed 15 users from various age groups, recording 10 characteristics and behaviors for each individual. This resulted in a total of 150 unique behavioral patterns. We analyzed and categorized these behaviors, creating four primary personas that represent the types of people engaging with this market.

From these personas, we selected two with the most contrasting behavioral traits as our primary design focus:

- Persona 1: a young design student who loves exploring the city
- Persona 2: an elderly fabric shop owner passionate about preserving the stories behind textiles.

This contrast allowed us to address diverse needs and perspectives, providing a foundation for crafting a meaningful and inclusive design strategy.

2.2 Pair Personas and AIGC Developing

Based on the primary and secondary personas derived from our field research, we employed **AIGC** (Artificial Intelligence-Generated Content) methods to further develop and refine the personas. By inputting the initial characteristics and behavioral patterns of the two personas into ChatGPT-4o, we instructed the AI to role-play as these individuals. Through in-depth interviews and conversations with the AI-generated personas, we explored their thoughts, motivations, and experiences in greater detail.

The insights gained from these interactions were then compared with and adjusted against the original persona descriptions, enabling us to enrich and deepen their behavioral characteristics. This iterative process allowed us to craft more nuanced and realistic personas, providing a robust foundation for our design strategy that addresses both their explicit and implicit needs within the context of the Ximen Asakusa Fabric Market.

2.3 User Journey Map and Design Direction

After refining the two distinct personas, we used their behavioral characteristics and insights from the AIGC-assisted interviews to create user journey maps that detailed their experiences within the Ximen Asakusa Fabric Market. By analyzing these maps, we identified overlapping touchpoints and areas needing improvement, which we categorized into three main design challenges and directions:

- Promotion and preservation of fabric culture and the fabric market.
- Improvement of the spatial flow within the market.
- Enhancement of the fabric purchasing experience.

In alignment with our initial focus on fabric culture and sustainable fashion, we selected the first challenge—the promotion and preservation of fabric culture and the market—as our primary design focus. This decision was driven by its potential to address both cultural and sustainability goals, creating a meaningful connection between tradition and contemporary urban life.

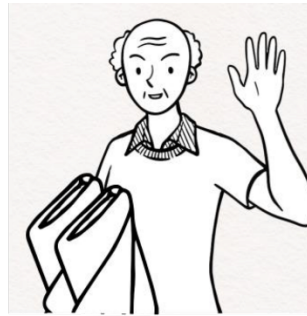


Figure 2: Persona 1 and 2 imageries

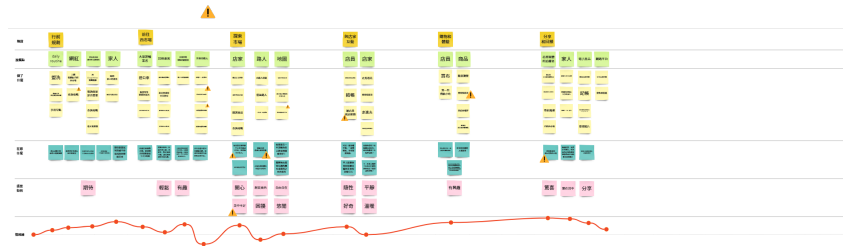


Figure 3: User Journey Map (Persona 1)

2.4 Design Strategy

For the design direction focusing on the promotion and preservation of fabric culture and the fabric market, we drew inspiration from the characteristics of the first persona—a young design student who loves exploring the city. Combining their passion for urban exploration with the concept of collecting, we developed the idea of Fabric Tag#, a concept and series of experiential designs.

This concept transforms fabric culture into a citywide experience, allowing participants to discover and interact with fabric-related stories and artifacts through exploration and collection. Our team envisions Fabric Tag# as a means to disseminate the essence of fabric culture throughout the urban landscape, offering both personas—the young design student and the elderly fabric shop owner—distinct yet interconnected experiences. This approach not only bridges tradition and innovation but also fosters a deeper connection between diverse stakeholders within the context of fabric culture.

3 Design Solution

3.1 What is Fabric Tag#?

The proposed solution consists of two integral components: Fabric Tag#, the physical tags, and the Fabric Tag# app. Fabric Tag# is crafted from various fabric types, offering flexibility in its use either independently or attached to other fabric-based products. Each tag features a sewn-in QR code, which, when scanned with a mobile device, links directly to the Fabric Tag# app. Functioning as unique identifiers for the fabric, these tags employ technology to document and disseminate comprehensive information regarding the material, craftsmanship, and distinctive narrative associated with each fabric. This integration of physical and digital elements enables a dynamic approach to preserving and sharing fabric culture, enriching the user experience and fostering deeper connections to textile heritage.

Next, we will explain the following three points: how to obtain the Fabric Tag#, the physical and virtual applications of the Fabric Tag#, and the integration of technology with Fabric Tag#.

3.2 How Fabric Tag# can be acquired

Our focus centers on the younger generation's enthusiasm for city exploration and collectibles (Primary Persona). Fabric Tag# is designed to be integrated throughout the urban landscape, serving as material labels for clothing and fabric products. These tags can be collected by users when they purchase fabric-related products, thus turning them into tangible symbols of both the products and the city's fabric culture.



Figure 4: Fabric Tag# and Fabric Tag# App

In addition, businesses and organizations could co-brand their own Fabric Tag# for promotional purposes, offering them as gifts or selling them as merchandise. Cultural events such as exhibitions or tours can distribute Fabric Tag# as souvenirs, encapsulating the essence of the experience. City landmarks and festivals can release limited-edition Fabric Tag# tags, each tailored to reflect the unique cultural characteristics of the city or the event, enhancing their appeal as collectible mementos.

Furthermore, Fabric Tag# can collaborate with leading fabric manufacturers, textile merchants, and fashion brands to promote sustainable practices and values. By fostering these partnerships, Fabric Tag# can extend its influence on a global scale, contributing to the growth of sustainable fashion and reinforcing its role in the preservation and celebration of fabric culture across diverse regions.

3.3 Fabric Tag# functions in both physical and virtual environments

3.3.1 Physical Uses of Fabric Tag#. The physical tags facilitate a tactile engagement with the fabric's texture, functioning both as a personal swatch book and as a decorative accessory. Furthermore, these tags promote social interaction by providing users with the opportunity to discuss, exchange, or trade them with others. This dynamic fosters the creation of a community centered around the Fabric Tag#, thereby enhancing both individual and collective experiences related to the material.

3.3.2 Digital Uses of Fabric Tag#. When users scan the QR code on the reverse side of a Fabric Tag# using their mobile device, they can register the tag within the online Fabric Tag# database. The accompanying app systematically organizes and tracks the user's tags, ensuring that fabric-related information remains accessible



Figure 5: Fabric Tag# as fabric labels and souvenirs, are scattered throughout the city for people to collect.



Figure 6: The use and exchange of physical Fabric Tag#.

even if the physical tag is lost or damaged. Each fabric's dedicated page within the app provides comprehensive details regarding its material composition, manufacturing process, history, narrative, and purchasing options. Additionally, the app fosters community engagement by enabling users to share their thoughts, creations, and experiences. Through its integrated social platform, users can interact with one another by liking, commenting on, and sharing posts, thereby cultivating a dynamic and interactive fabric-centered community.

3.4 Fabric Tag# connects with and utilizes technology.

3.4.1 For Public Users (Persona 1). Serving as the conduit between the physical and virtual realms of Fabric Tag#, the app incorporates interactive features enhanced by cutting-edge technology. The Augmented Reality (AR) Mode allows users to visualize various fabric textures and designs on everyday items, such as sofas or backpacks, providing a realistic simulation of their appearance in the physical environment. The Free Design feature empowers users to personalize products by utilizing in-app templates and selecting their preferred fabrics, facilitating the creation of bespoke, individualized items. Furthermore, the app enables users to explore local markets or stores in Tainan for additional customization options or to engage in workshops, thereby transforming their creative concepts into tangible outcomes.

3.4.2 Support for Local Fabric Retailers from Fabric Tag# (Persona 2). The design of Fabric Tag# integrates features that support local fabric retailers by assisting in the formulation of marketing strategies and the introduction of new products, informed by market trend insights derived from the app's data analytics. Retailers can also optimize inventory management through the app's data center, which monitors prices, stock levels, and sales performance. Additionally, the app offers retailers a platform to share their crafting techniques, and the cultural narratives associated with their fabrics, ensuring the preservation and widespread dissemination of these rich stories. Through Fabric Tag#, we seek to inspire users to reconsider the role of fabric in their lives, elevating it from a mere material to a medium for expressing cultural values.

4 Conclusion

Through the design of FABRIC TAG#, we aim to encourage users to reconsider the role of fabric in their lives. Fabric will no longer be perceived merely as a material, but rather as a medium for conveying cultural values. By focusing on three key design dimensions—How Fabric Tag# can be acquired, its functions in both physical and virtual environments, and its integration with technology—we strive to achieve transformative changes in three distinct areas:

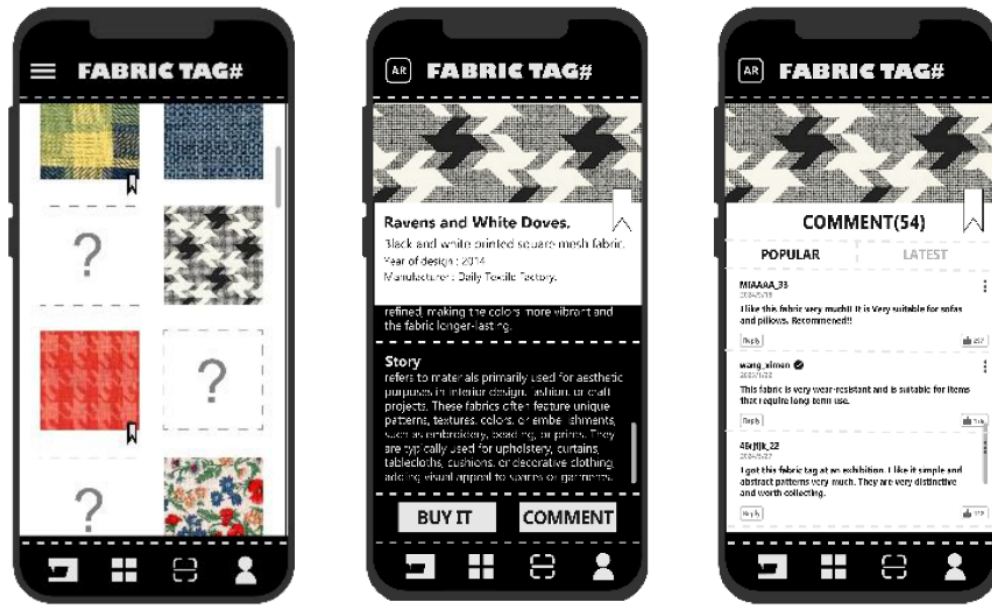


Figure 7: The screenshots of Collection Book, Story Book, and Social Platform in Fabric Tag# app.

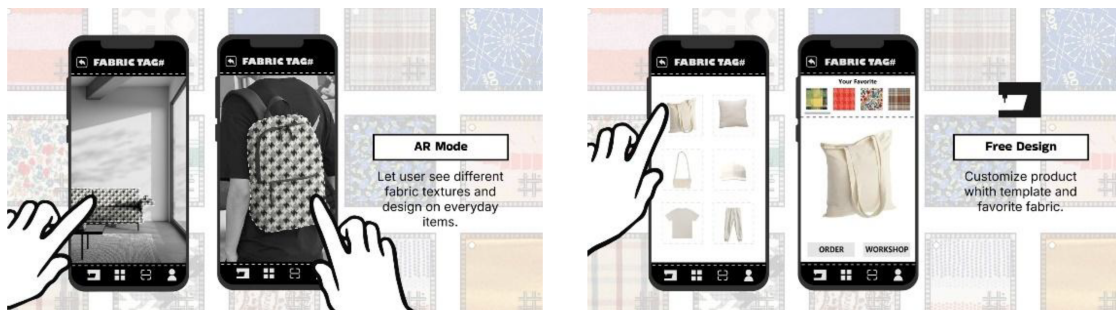


Figure 8: The screenshots of AR Mode, Free Design, and Course Redemption in Fabric Tag# app.



Figure 9: The screenshots of Fabric Tag# Data Center in the app.

4.1 Weaving Culture into the Cityscape: Rediscovering Fabric Through Urban Exploration:

This experience design reintroduces fabric into people's daily lives through urban exploration, while simultaneously emphasizing its cultural significance and its deep connection to the history of the city.

4.2 Fabric Tags: Bridging Disciplines to Build Sustainable Urban Communities:

The physical and virtual Fabric Tags will serve as a unifying force, bringing together individuals from diverse fields and professions. This fusion will foster the creation of new urban communities and stimulate discussions on fabric, fashion, and sustainability.

4.3 Fusing Technology and Sustainability: Reviving Craftsmanship and Challenging Fast Fashion

Our objective is to merge technology with sustainable development, reigniting a culture that appreciates and values custom-made creations [14]. The platform invites users to critically reconsider the environmental challenges posed by fast fashion.

Ultimately, through the experience design of Fabric Tag#, we hope that fabric will evolve into a new form, becoming an integral part of our everyday lives. It will serve as a carrier of cultural heritage, a medium for urban exchange, and a catalyst for sustainable fashion. Let the beauty of fabric once again be seen and appreciated.

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References

- [1] G. PANKAJ and N. JOSÉ LUIS, "Zara Fast Fashion Case Study." Dec. 21, 2006.
- [2] V. Bhardwaj and A. Fairhurst, "Fast fashion: response to changes in the fashion industry," *Int. Rev. Retail Distrib. Consum. Res.*, vol. 20, no. 1, pp. 165–173, Feb. 2010, doi: 10.1080/09593960903498300.
- [3] "The slow burn of fast fashion: Unsustainable practices within the industry's supply chain." Accessed: Jan. 22, 2025. [Online]. Available: <https://www.usf.edu/business/documents/undergraduate/honors/thesis-payne-katelyn.pdf>
- [4] K. Niinimäki, G. Peters, H. Dahlbo, P. Perry, T. Rissanen, and A. Gwilt, "The environmental price of fast fashion," *Nat. Rev. Earth Environ.*, vol. 1, no. 4, pp. 189–200, Apr. 2020, doi: 10.1038/s43017-020-0039-9.
- [5] E. Paulicelli and H. Clark, *Fabric of Cultures*. Routledge, 2008. Accessed: Jan. 22, 2025. [Online]. Available: [https://api.taylorfrancis.com/content/books/mono/download?identifierName=\\$doi&identifierValue=\\$10.4324/9780203869024&type=\\$googlepdf](https://api.taylorfrancis.com/content/books/mono/download?identifierName=$doi&identifierValue=$10.4324/9780203869024&type=$googlepdf)
- [6] "Material of culture, fabric of identity," in *Material Cultures*, Routledge, 2002, pp. 121–145. doi: 10.4324/9780203033142-9.
- [7] W. M. Reddy, *The Rise of Market Culture: The Textile Trade and French Society, 1750–1900*. Cambridge University Press, 1987.
- [8] M. Aakko and R. Koskenmuri-Sivonen, "Designing Sustainable Fashion: Possibilities and Challenges," *Res. J. Text. Appar.*, vol. 17, no. 1, pp. 13–22, Feb. 2013, doi: 10.1108/RJTA-17-01-2013-B002.
- [9] C.-C. Hu and C.-C. Hu, "The Socioeconomic-Spatial Transformations of Tainan City (1875–2003): A Case of Textile and Garment Industries," thesis, 2018. Accessed: Jan. 22, 2025. [Online]. Available: <https://nckur.lib.ncku.edu.tw/handle/987654321/260213>
- [10] Z.-S. Wang, H.-K. Duan, and F.-J. Zeng, *Distant Weaving Paths: The Development and Evolution of the Textile Industry in Tainan over the Past 400 Years*, 1st ed., vol. 1, 1 vols. Taiwan: Cultural Affairs Bureau, Tainan City Government., 2022.
- [11] SI-MIN QIAN, "Digital transformation drives Taiwan's industry and software overseas opportunities - taking the textile industry as an example," *Taiwan Econ. Res. Mon.*, vol. 47, no. 6, pp. 48–55, Jun. 2024, doi: 10.29656/TERM.202406_47(6).0007.
- [12] S.-Y. WU, "Research on the Enterprise Transformation of Taiwan Textile Industry," *Masters Thesis Natl. Taipei Univ. Bus.*, pp. 1–106, Jan. 2018.
- [13] W.-H. Lin and W.-H. Lin, "Refashion: Clothing Mending as a Regeneration System of Tainan's Fabric Market," thesis, 2016. Accessed: Jan. 22, 2025. [Online]. Available: <https://nckur.lib.ncku.edu.tw/handle/987654321/260326>
- [14] S. Jinhui Wu, D. Zhang, and R. G. Schroeder, "Customization of quality practices: the impact of quality culture," *Int. J. Qual. Reliab. Manag.*, vol. 28, no. 3, pp. 263–279, Jan. 2011, doi: 10.1108/026567111111109883.